

Dissertation

A Distributed Edge FLISR Solution & Network Simulation Test Platform

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1 Introduction

Similar style and content as the proposal document, possibly expand it out where appropriate.

1.1 Research background

1.2 Motivations for the research

1.3 Research objectives

2 Review of the literature

Similar style and content as the proposal document, expand out into more research topics (bring in simulations?)

2.1 Digitalisation of the grid

2.2 Fault Location, Isolation & Service Restoration

2.3 Edge computing

3 Methods

Methods section will describe the context of the research and the actual work done throughout the dissertation.

3.1 Trial grid topology features

Describe the trial sites selected to use for simulations and testing, describe the features (network structure, no. of switches etc.)

3.2 FLISR Edge solution & algorithm

Describe the FLISR Edge solution, how it is structured, orchestrated, the edge devices, algorithms etc.

3.3 Network simulation test platform

Describe simulation platform and its structure, purpose, how it operates etc.

3.4 Fault simulation operation

Describe how the FLISR Edge solution and simulation platform interact and operate, how it generates results and how these are presented.

4 Results of the research

Section describing the results gathered from the testing, including the findings, relevance of the results to the DSO and consumer, and future work to expand the solution further.

4.1 Research findings

Describe the findings, how effective the solution is (or not), compare and contrast across the selected trial grids, centralised vs. distributed, limitations etc.,

4.2 Relevance to the DSO

Describe the relevance of the results to the DSO and the consumer, from a grid stability, financial and climate perspective (weather events).

4.3 Future work

5 Conclusion

6 Data & software availability

Describe how the software and background data can be accessed (is this needed/relevant?)

References